

SAHIL SHAH

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RESEARCH AND PROFESSIONAL EXPERIENCE.

University of Maryland

Assistant Professor

Spring 21 -Present

College Park

My areas of research are low-power analog and mixed-signal systems for energy-efficient computation, embedded machine-learning and bio-sensing and continuous monitoring devices. Long-term goal of my work is to investigate energy-efficient solid-state circuits and architecture to compute and process signals in real-time.

California Institute Of Technology

Postdoctoral Associate

July 18 - December 20

Pasadena

- My work involved designing hardware and algorithm for brain-machine interface. Eventual goal of this work is to develop an implantable system which could control prosthetics. Specific focus of this work involves designing decoders for mapping neural spike data to kinematics for controlling cursor on the computer. The neural data is recorded from electrodes (UTAH NeuroPort) implanted in supramarginal gyrus (SMG), ventral premotor cortex (PMV), and primary somatosensory cortex (S1). We are currently testing the neural network decoders with the human subject in the loop. I am also investigating methods to design energy-efficient neural network architectures on FPGA and fabricated it on a 65nm ASIC. I am also investigating analog neural networks, with devices biased in sub-threshold regime, for further reducing the power consumption.

Georgia Institute Of Technology

Graduate Research Assistant

Fall 14 - May 18

Atlanta

- I worked on designing a mixed-signal Floating-Gate based FPAA (Field Programmable Analog Array) in 130nm technology. I also used the FPAA for performing energy-efficient computation for processing signals from different sensors. This system was also used to process physiological signals for monitoring vital signals of patients. I also worked on implementing analog neural network for command word recognition on the FPAA. Such systems could be used for context aware wearable devices where low-power consumption is important.

Arizona State University

Graduate Research Assistant

Fall 11 - Spring 14

Tempe

- I worked on designing read-out circuits for CMOS based bio-sensors in 0.5um technology. Specifically, to measure change in pH of the cell culture media. I also developed bio-sensing platform using flexible electronics (originally used for flexible displays) for developing smart skin.

EDUCATION

Georgia Institute of Technology

Ph.D in Electrical Engineering

August 2014 - March 18

Thesis: Low-Power Computation Using FPAA for Wearable Devices.

PI: Dr. Jennifer Hasler

Arizona State University

M.S in Electrical and Computer Engineering

Fall 2011 - Spring 2014

Thesis: Biosensors and CMOS Interface Circuits.

PI: Dr. Jennifer Blain Christen

AWARDS

Best Associate Editor for IEEE Transactions on Biomedical Circuits and Systems 2025-2026

Best Associate Editor for IEEE Open Journal of Circuits and Systems 2024-2025

NSF CAREER Award (2024)

George Corcoran Memorial Award for Significant Contributions to Electrical and Computer Engineering Education (2023)
Received Best Design Award by Analog Devices Inc. at International Solid-State Circuits Conference (ISSCC) (2015)
Marion A. and Henry C. Bourne Fellowship for attending Georgia Institute of Technology (2014)

PUBLICATIONS: JOURNALS

1. Sayma Nowshin Chowdhury , Alex L. Mazzoni, Xiaohang Zhang, Andreu L. Glasmann, Halid Mulaosmanovic, Stefan D unkel, Gunda Beernink, Sven Beyer, Sina Najmaei, and Sahil Shah "Measurement and Analysis of Multistate Ferroelectric Transistors in 28 nm CMOS Process," in IEEE Journal of the Electron Devices Society 2026, doi: 10.1109/JEDS.2026.3664683.
2. Irem Didin, Carl Brando, Ching-Yi Lin and Sahil Shah "Characterization and Modeling of Multilevel Analog ReRAM Synapses in the Sky130 Process," in IEEE Journal on Exploratory Solid-State Computational Devices and Circuits, doi: 10.1109/JXCDC.2026.3670667.
3. Yongmin Baek, Byungjoon Bae, Hyojin Shin, Charana Sonnadara, Haein Cho, Ching-Yi Lin, Yujia Mu, Cong Shen, Sahil Shah, Gunuk Wang, Kyusang Lee "Edge intelligence through in-sensor and near-sensor computing for the artificial intelligence of things" *npj Unconventional Computing*, vol. 2, Article 25 (2025).
4. Charana Sonnadara and Sahil Shah "Real-time Analog Processing with On-Chip Learning using Multiple-Input Translinear Elements" npj Unconv. Comput. 2, 11 (2025).
5. Mudi Zhang, Charana Sonnadara, Sahil Shah, and Min Wu "Feasibility Study of Location Authentication for IoT Data Using Power Grid Signatures" in IEEE Open Journal of Signal Processing, vol. 6, pp. 405-416, 2025.
6. Sayma Nowshin Chowdhury and Sahil Shah "Analysis and Verilog-A Modeling of Floating-Gate Transistors" in IEEE Open Journal of Circuits and Systems, vol. 6, pp. 63-73, 2025.
7. Matthew Chen, Charana Sonnadara, and Sahil Shah "Open-source Floating-Gate Cell for Analog Synapses" IET Electronic Letters 2024.
8. Elijah Taeckens and Sahil Shah "A Spiking Neural Network with Continuous Local Learning for Robust Online Brain-Machine Interface" Journal of Neural Engineering January 2024 doi: 10.1088/1741-2552/ad1787
9. Byungjoon Bae, Yongmin Baek, Jeongyong Yang, Heesung Lee, Charana S. S. Sonnadara, Sangeun Jung, Minseong Park, Doeon Lee, Sihwan Kim, Gaurav Giri, Sahil Shah*, Geonwook Yoo*, William A. Petri*, and Kyusang Lee* "Near-sensor computing-assisted simultaneous viral antigen and antibody detection via integrated label-free biosensors with microfluidics" InfoMat.2023;e12471. <https://doi.org/10.1002/inf2.12471> *-Corresponding Authors
10. H. Yang, L. Jia, X. Xiao, J. Wang, Y. Li, K. Li, Z. Li, J. Yang, J. S. Ho, P.-L. Yeh, K. Mouthaan, X. Wang, Sahil Shah*, and P.-Y. Chen*, "Topographic Design in Wearable MXene Sensors with In-Sensor Machine Learning for Full-Body Avatar Reconstruction," Nature Communication 13, 5311 (2022).*Corresponding Authors
11. Sihwan Kim, Sahil Shah, Richard Wunderlich, and Jennifer Hasler "CAD Synthesis Tools for Floating-Gate SoC FPAAs" Design Automation for Embedded Systems March 2021.
12. Jennifer Hasler and Sahil Shah "An SoC FPAA-Based Programmable, Ladder-Filter-Based, Linear-Phase Analog Filter" in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 68, no. 2, pp. 592-602, Feb. 2021.
13. Sahil Shah, Hakan Toreyin, Gungor Cih as, and Jennifer Hasler "A Real-Time Vital-Sign Monitoring in the Physical Domain on a Mixed-Signal Reconfigurable Platform" in IEEE Transactions on Biomedical Circuits and Systems, vol. 13, no. 6, pp. 1690-1699, Dec. 2019 doi: 10.1109/TBCAS.2019.2949778.

14. Sahil Shah and Jennifer Hasler “SoC FPAA Hardware Implementation of a VMM+ WTA Embedded Learning Classifier” IEEE Journal on Emerging and Selected Topics in Circuits and Systems November vol. 8, no. 1, pp. 28-37, March 2018.
15. Sahil Shah; H. Toreyin; J. Hasler; and A. Natarajan “Temperature Sensitivity and Compensation on a Reconfigurable Platform” IEEE Transactions on Very Large Scale Integration (VLSI) Systems briefs vol. 26, no. 3, pp. 604-607, March 2018.
16. Jennifer Hasler; Sahil Shah “VMM + WTA Embedded Classifiers Learning Algorithm Implementable on SoC FPAA Devices” in IEEE Journal on Emerging and Selected Topics in Circuits and Systems, vol. 8, no. 1, pp. 65-76, March 2018.
17. Sahil Shah; Hakan Toreyin; Jennifer Hasler; and Aishwarya Natarajan “Models and Techniques for Temperature-Robust Systems on a Reconfigurable Platform” Journal of Low Power Electronics and Applications 2017, 7(3), 21.
18. Sahil Shah; J. Hasler “Tuning of Multiple Parameters with a BIST System” in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 64, no. 7, pp. 1772-1780, July 2017.
19. S. Kim; Sahil Shah; and J. Hasler “Calibration of Floating-Gate SoC FPAA System” IEEE Transactions on Very Large Scale Integration (VLSI) Systems vol. 25, no. 9, pp. 2649-2657, Sept. 2017.
20. George, S.; Kim, S.; Sahil Shah, et al. “A Programmable and Configurable Mixed-Mode FPAA SoC” IEEE Transactions on Very Large Scale Integration (VLSI) Systems vol. 24, no. 6, pp. 2253-2261, June 2016.
21. Hasler, J.; Sahil Shah; S. Kim; Kumal Lal, I.; and Collins, M. (2016), “Remote System Setup Using Large-Scale Field Programmable Analog Arrays (FPAA) to Enabling Wide Accessibility of Configurable Devices” J. Low Power Electron. Appl. 6(3), 14.
22. Sahil Shah; Smith, J.; Stowell, J.; and Christen, J. B. (2015), “Biosensing Platform on a Flexible Substrate” Sensors and Actuators B: Chemical 210, 197–203.
23. Smith, J. T.; Sahil Shah; Goryll, M.; Stowell, J. R.; and Allee, D. R. (2014), “Flexible ISFET Biosensor Using IGZO Metal Oxide TFTs and an ITO Sensing Layer” IEEE Sensors Journal 14(4), 937–938.
24. Welch, D.; Sahil Shah; Ozev, S.; and Christen, J. B. (2013), “Experimental and Simulated Cycling of ISFET Electric Fields for Drift Reset” IEEE Electron Device Letters 34(3), 456–458.

PUBLICATIONS: REFERRED PEER REVIEWED CONFERENCES

1. Ching-Yi Lin and Sahil Shah, ”Design Space Exploration for ReRAM-based Architectures to Address Scaling Non-idealities,” Accepted to IEEE International Midwest Symposium on Circuits and Systems (MWSCAS).
2. Yihui Wang, Sheng-Yu Peng and Sahil Shah, ”An Ultra-Low-Power Synthesizable Asynchronous AER Encoder for Neuromorphic Edge Devices,” Accepted to IEEE International Midwest Symposium on Circuits and Systems (MWSCAS).
3. Kaushik Lakshmiramanan, Vineeta Nair, Ching-Yi Lin, Sheng-Yu Peng and Sahil Shah, ”An Asynchronous Delta Modulator for Spike Encoding in Event-Driven Brain-Machine Interface,” Accepted to IEEE International Midwest Symposium on Circuits and Systems (MWSCAS).
4. Everest Bloomer, Irem Didin, Ching-Yi Lin and Sahil Shah, ”Ising-ReRAM: A Low Power Ising Machine ReRAM Crossbar for NP-Complete Problems,” 2026 IEEE International Symposium on Circuits and Systems (ISCAS), Shanghai, China, 2026.
5. Vineeta V Nair, Sayma Nowshin Chowdhury and Sahil Shah, ”Variation-Tolerant Hardware synthesis of Spiking neural networks,” 2025 ICedge.

6. Ching-Yi Lin and Sahil Shah, "Low-Bit Integerization of Vision Transformers using Operand Reordering for Efficient Hardware," 2025 IEEE International Midwest Symposium on Circuits and Systems (MWSCAS).
7. U. Noyan, Sahil Shah and P. Abshire, "Functionalized ImmunoFET for Detection of Phosphatidyl-L-serine," 2025 IEEE International Symposium on Circuits and Systems (ISCAS), London, United Kingdom, 2025, pp. 1-5.
8. C. Sonnadara and Sahil Shah, "Verilog-A modeling of Floating-Gate (FG)-based Multiple-Input Translinear Elements," 2025 IEEE International Symposium on Circuits and Systems (ISCAS), London, United Kingdom, 2025, pp. 1-5.
9. Jeremy Yun, Steeve Nzama, Sahil Shah, "Towards a Broadly Configurable Wearable Device for Continuous Hemodynamic Monitoring," 2023 IEEE 67th International Midwest Symposium on Circuits and Systems (MWSCAS), Springfield, MA, USA, 2024, pp. 1-5.
10. C. Sonnadara and Sahil Shah, "On-Chip Adaptation for Reducing Mismatch in Analog Non-Volatile Device-Based Neural Networks," 2024 IEEE International Symposium on Circuits and Systems (ISCAS), Singapore, Singapore, 2024, pp. 1-5, doi: 10.1109/ISCAS58744.2024.10557839.
11. Utku Noyan, Sahil Shah, and Pamela Abshire, "Drift Behavior Analysis of ISFET Models Using COMSOL Multiphysics," 2024 IEEE Biosensors Conference (BioSensors), Cambridge, UK, 2024.
12. Utku Noyan, Sheung Lu, Jennifer Blain Christen, Pamela Abshire, and Sahil Shah, "Scalable Biosensors Using Standard CMOS Process," 2023 IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, USA, 2023.
13. Krishna, Adithya, Ramanathan, Vignesh, Yadav, Satyapreet Singh, Sahil Shah, van Schaik, André, Mehendale, Mahesh, and Thakur, Chetan Singh, "A Sparsity-Driven tinyML Accelerator for Decoding Hand Kinematics in Brain-Computer Interfaces," 2023 IEEE Biomedical Circuits and Systems Conference (BioCAS), Toronto, ON, Canada, 2023, pp. 1-5, doi: 10.1109/BioCAS58349.2023.10389094.
14. Sayma Nowshin Chowdhury and Sahil Shah, "Design Space Exploration Tool for Mixed-Signal Spiking Neural Network," 2023 IEEE 66th International Midwest Symposium on Circuits and Systems (MWSCAS), Phoenix, AZ, USA, 2023, pp. 1-5.
15. Utku Noyan, Glenn Ray, Avery Snow Cobb, Pamela Abshire, and Sahil Shah, "COMSOL Modeling of ISFET for pH Sensing and Antigen-Antibody Detection," 2023 IEEE 66th International Midwest Symposium on Circuits and Systems (MWSCAS), Phoenix, AZ, USA, 2023, pp. 1-5.
16. Charana Sonnadara, Mudi Zhang, Min Wu, and Sahil Shah, "Low-Power Mixed-Signal System for Processing Electric Network Frequency in IoT Devices," 2023 IEEE 66th International Midwest Symposium on Circuits and Systems (MWSCAS), Phoenix, AZ, USA, 2023, pp. 1-5.
17. Carl Brando, Minseong Park, Sayma Nowshin Chowdhury, Matthew Chen, Kyusang Lee, and Sahil Shah, "Modeling and Analysis of Analog Non-Volatile Devices for Compute-In-Memory Applications," 2023 IEEE 66th International Midwest Symposium on Circuits and Systems (MWSCAS), Phoenix, AZ, USA, 2023, pp. 1-5.
18. E. Taeckens, R. Dong, and Sahil Shah, "A Biologically Plausible Spiking Neural Network for Decoding Kinematics in the Hippocampus and Premotor Cortex," 2023 11th International IEEE/EMBS Conference on Neural Engineering (NER), Baltimore, MD, USA, 2023, pp. 1-4.
19. Sahil Shah, Hakan Toreyin, Utku Noyan, and Yoo Jin Lee, "A Proof-of-Concept Real-Time Processing to Characterize Vascular Flow," 2022 44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Glasgow, UK, 2022, pp. 2912-2915.
20. Sayma Nowshin Chowdhury and Sahil Shah, "Hardware-Aware Modeling of Mixed-Signal Spiking Neural Network," 2022 20th IEEE Interregional NEWCAS Conference (NEWCAS), Quebec City, QC, Canada, 2022, pp. 104-108, doi: 10.1109/NEWCAS52662.2022.9842116.

21. Sahil Shah and J. B. Christen, "Special Session: Calibrating Mismatch in an ISFET with a Floating-Gate," *2022 IEEE 40th VLSI Test Symposium (VTS)*, San Diego, CA, USA, 2022, pp. 1-4.
22. Benyamin Haghi, Spencer Kellis, Sahil Shah, Maitreyi Ashok, Luke Bashford, Daniel Kramer, Brian Lee, Charles Liu, Richard A. Andersen, and Azita Emami, "Deep Multi-State Dynamic Recurrent Neural Networks Operating on Wavelet-Based Neural Features for Robust Brain-Machine Interfaces," *Conference on Neural Information Processing Systems (NeurIPS)*, Vancouver, BC, Canada, 2019.
23. Sahil Shah, Benyamin A. Haghi, Spencer Kellis, Luke Bashford, Daniel Kramer, Brian Lee, Charles Liu, Richard A. Andersen, and Azita Emami, "Decoding Kinematics from Human Parietal Cortex Using Neural Networks," *IEEE Brain Initiative Workshop on Advanced NeuroTechnologies*, San Francisco, CA, USA, 2019, pp. 1-4.
24. Jennifer Hasler and Sahil Shah, "Enabling Embedded Learning and Classification Implemented on SoC FPAA Devices," *2018 IEEE International Symposium on Circuits and Systems (ISCAS)*, Florence, Italy, 2018, pp. 1-5.
25. Sahil Shah and Jennifer Hasler, "Low Power Speech Detector on an FPAA," *2017 IEEE International Symposium on Circuits and Systems (ISCAS)*, Baltimore, MD, USA, 2017, pp. 1-4.
26. Sihwan Kim, Sahil Shah, and Jennifer Hasler, "Floating-Gate FPAA Calibration for Analog System Design and Built-in Self-Test," *2017 IEEE International Symposium on Circuits and Systems (ISCAS)*, Baltimore, MD, USA, 2017, pp. 1-4.
27. Jennifer Hasler, Aishwarya Natarajan, and Sahil Shah, "SoC FPAA Immersed Junior Level Circuits Course," *2017 IEEE International Conference on Microelectronic Systems Education (MSE)*, Lake Louise, AB, Canada, 2017, pp. 7-10.
28. Hakan Toreyin, Sahil Shah, Sinan Hersek, Omer Inan, and Jennifer Hasler, "Proof-of-Concept Energy-Efficient and Real-Time Hemodynamic Feature Extraction from Bioimpedance Signals Using a Mixed-Signal Field Programmable Analog Array," *2017 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, Orlando, FL, USA, 2017, pp. 233-236.
29. Michelle Collins, Jennifer Hasler, and Sahil Shah, "An Approach to Using RASP Tools in Analog Systems Education," *2016 IEEE Frontiers in Education Conference (FIE)*, Erie, PA, USA, 2016, pp. 1-8.
30. Jennifer Hasler, Sihwan Kim, Sahil Shah, Farhan Adil, Michelle Collins, Scott Koziol, and Stephen Nease, "Transforming Mixed-Signal Circuits Class Through SoC FPAA IC, PCB, and Toolset," *11th European Workshop on Microelectronics Education (EWME)*, 2016, pp. 1-6.
31. Mahalanabis, D., Sivaraj, M., Chen, W., Sahil Shah, Barnaby, H. J., Kozicki, M. N., Christen, J. B., and Vrudhula, S., "Demonstration of Spike Timing Dependent Plasticity in CBRAM Devices with Silicon Neurons," *2016 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2016, pp. 2314-2317.
32. Sahil Shah, Jennifer Hasler, Sihwan Kim, Ishan Lal, Matt Kagle, and Michelle Collins, "A Remote FPAA System for Research and Education," *2016 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2016.
33. Sahil Shah, Hakan Toreyin, Omer T. Inan, and Jennifer Hasler, "Reconfigurable Analog Classifier for Knee-Joint Rehabilitation," *2016 IEEE 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2016, pp. 4784-4787.
34. Sahil Shah, Caitlin N. Teague, Omer Inan, and Jennifer Hasler, "A Proof-of-Concept Classifier for Acoustic Signals from the Knee Joint on an FPAA," *2016 IEEE Sensors Conference*, 2016, pp. 1-3.
35. Hany Arafa, Sahil Shah, and Christen, J. B., "An Integrated System for Continuous Assessment of Inter-cellular pH," *2014 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, 2014, pp. 332-335.

36. Sahil Shah, Anderson, K. S., Blain Christen, J., and Hasler, J., “Floating Gate ISFET for Therapeutic Drug Screening of Breast Cancer Cells,” *2014 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2014, pp. 229–232.
37. Sahil Shah, Arafa, H., and Christen, J. B., “An On-Chip System to Monitor the pH of Cell Culture Media,” *2014 IEEE Engineering in Medicine and Biology Society (EMBC)*, 2014, pp. 2745–2748.
38. Smith, J. T., Katchman, B. A., Lee, Y.-K., O’Brien, B. P., Bawolek, E. J., Sahil Shah, and Christen, J. B., “Disposable Point-of-Use Optical Biosensor for Multiple Biomarker Detection,” *2014 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, 2014, pp. 268–271.
39. Sahil Shah and Christen, J. B., “Pulse Width Modulation Circuit for ISFET Drift Reset,” *2013 IEEE Sensors Conference*, 2013, pp. 1–4.
40. Bolte, B., Sahil Shah, Kim, S., Hwang, P., and Hasler, J., “Live Demonstration: FPAA Demonstration Controlled Through Android-Based Device,” *2016 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2016, pp. 1442–1442.
41. Sahil Shah, Hasler, J., Kim, S., Lal, I., Kagle, M., and Collins, M., “Demonstration of a Remote FPAA System for Research and Education,” *2016 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2016, pp. 1441–1441.
42. Kim, S., Adil, F., Koziol, S., Nease, S., Collins, M., Sahil Shah, Kagle, M., and Hasler, J., “SoC FPAA IC, PCB, and Tool Demonstration,” *2016 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2016, pp. 1440–1440.
43. J. Hasler and Sahil Shah, “Reconfigurable Analog PDE Computation RF and Baseband Computation,” *GOMAC 2017*.
44. J. Hasler and Sahil Shah, “Learning for VMM + WTA Embedded Classifiers,” *GOMAC 2016*.
45. Sahil Shah, S. Kim, F. Adil, J. Hasler, S. George, M. Collins, R. Wunderlich, S. Nease, S. Ramakrishnan, “A Programmable and Configurable Mixed-Mode FPAA SoC,” *GOMAC 2016*.

PATENT: GRANTED AND PENDING APPLICATIONS

Smith, T. J.; Goryll, M.; Sahil Shah; Blain Christen, J.; and Stowell, J. (2015), “System and Method for Ion-Selective Field Effect Transistor on Flexible Substrate” US Patent US10249741B2, Filing Date: 05-13-2015.

Jeremy Yun and Sahil Shah ”Programmable Wearable Device for Hemodynamic Monitoring” Applied

TALKS AND WORKSHOPS

Invited talk at LockHeed Martin’s US-Intelligent Computing at the Edge Group 2026

Invited talk at TxACE UTdallas for SRC 2025 [Link to the Talk](#)

Invited talk at Army Research Lab 2024

Invited talk at Synopsys Armenia 2024

Invited talk at University of Delaware 2024

Invited talk at Northtop Grummann Symposium (NGS) 2023

Talk as part of Brain, Computation and Learning Workshop at Indian Institute of Science (IISC) 2023 [Link to the Workshop](#)

Virtual Tutorial at Indian Institute of Science (IISC), Bangalore on ”Analog Synapses for Computation” March 10th 2022.

Virtual Talk at Michigan State University, March 3rd 2022, [Link to the Talk](#)

Virtual Talk at University of Utah, Salt lake city, April 2020, [Link to the Talk](#)

Invited Talk at University of Texas at Dallas, Dallas, March 2020

Invited Talk at Columbia University, NewYork, May 2018

Invited Talk at San-Diego State University, San Diego, April 2018

Invited Talk at University of California Irvine, Irvine, April 2018

Invited Talk at University of Southern California, Los Angeles, April 2018

Mayo clinic: Demonstration of speech processing (Atlanta, August 2017)

Draper Labs: Demonstration of analog computation (Boston, December 2016)

Northrop Grumman: Demonstration on the use of FPAA (Washington DC, December 2014)

EDITORIAL ACTIVITIES

NSF review panels: NSF/ECCS/ERI, EFRI, NSF/SCH.

Associate Editor for IEEE Open Journal of Circuits and System 2024-Present.

Associate Editor for IEEE Transaction of Biomedical Circuits and Systems 2024-Present.

IEEE BioCAS Technical Program Committee. 2023-Present

ACM Design Automation Conference (DAC) Technical Program Committee. 2026-Present

IEEE CICC Technical Program Committee. 2024-Present

Reviewer for Transaction on Circuits and System (TCAS), Transaction on Biomedical Circuits and System (TBio-CAS), Journal of Biomedical Health and Informatics (JBHI), Journal of Solid-State Circuits (JSSC), IEEE Sensors journal, NEURIPS, MDPI Electronics, MDPI MEMS, MDPI technologies.

Mid-West Symposium on Circuits and System (MWSCAS) 2023 Program Committee